

(*) **SUMIFS()**:- It adds values that meet multiple conditions.

Syntax: =SUMIFS (sum-range, criteria-range1, criteria1, criteria-range2, criteria2, ...)



Example: Suppose:

Employee	Department	Salary
Ravi	IT	30000
Anu	HR	25000
Kiran	IT	35000
Roh	HR	28000

Total Salary of IT employees whose salary is greater than 30K :-

= SUMIFS (C2:C5, B2:B5, "IT", C2:C5, ">30000")
o/p :- 35,000

(*) AVERAGEIFS: This calculates the average of values that meet multiple conditions.

Syntax: = AVERAGEIFS (average_range, criteria_range1, criteria1, criteria_range2, criteria2, ...)

Example: Average salary of IT employees earning more than 30K :-

= AVERAGEIFS (C2:C5, B2:B5, "IT", C2:C5, ">30000")
o/p :- 35,000

(*) COUNTIFS: It counts the number of cells / records that satisfy multiple conditions.

Syntax: = COUNTIFS (criteria_range1, criteria1, criteria_range2, criteria2, ...)

Example: Count IT employees earning more than 30000.

= COUNTIFS (B2:B5, "IT", C2:C5, ">30000")
o/p :- 1

(*) IF: This checks one condition & returns one value if TRUE & condition another value if FALSE.

Syntax: $=IF(\text{logical_test}, \text{value_if_true}, \text{value_if_false})$

Example: If salary is 30K or more, display High otherwise Low.

$=IF(C2 \geq 30000, "High", "Low")$

O/p: LOW

(*) IF with AND: AND checks whether all conditions are TRUE. It is useful when you want to check 2 or more conditions together.

Syntax: $=IF(AND(\text{condition1}, \text{condition2}), \text{value_if_true}, \text{value_if_false})$

Example: If marks are at least 40 in both subjects.

$=IF(AND(B2 \geq 40, C2 \geq 40), "Pass", "Fail")$

(*) IF with OR: OR checks whether at least one condition is TRUE.

Syntax: $=IF(OR(\text{condition1}, \text{condition2}), \text{value_if_true}, \text{value_if_false})$

Example: If an employee belongs to IT or HR, display "Company Employee".

$=IF(OR(B2="IT", B2="HR"), "Company Employee", "Other")$

(*) IFS: IFS checks multiple conditions & returns the result corresponding to the first TRUE condition.

It is useful when you have several ranges/categories & want to avoid using many nested IFS.



Syntax: = IFS (condition1, result1, condition2, result2, condition3, result3, ...)

Example: = IFS (A2 <= 5000, 80, A2 <= 10000, 100, A2 <= 15000, 200, A2 <= 20000, 300, A2 > 20000, 400)

(*) COUNTIF:- Counts the number of cells that meet a single condition.

Syntax: COUNTIF (range, criterion).

Example: Count of employees in IT department

= COUNTIF(B2:B5, "IT")

o/p 1-2

dept	IT	HR	IT	HR
	(B2)	(B3)	(B4)	(B5)